

Alejandro Sánchez

github.com/aesbetancourt
Barcelona, Spain

linkedin.com/in/aesbetancourt

aesbetancourt@outlook.com

Summary

With a deep foundation in mathematics and software engineering, I specialize in architecting highly scalable, distributed systems using TypeScript, JavaScript, and Python. My expertise lies in high-level system design, performance optimization, and solving complex business logic that standard tools cannot. I view modern software development as an exercise in AI-assisted engineering and architectural orchestration. I leverage technical expertise to guide AI coding agents, rigorously audit generated code for structural integrity, and securely integrate LLMs into enterprise environments. By prioritizing the shift-left model—embedding automated testing, strict typing, and security from day one—I ensure that AI-accelerated workflows produce resilient, future-proof products built to scale.

Experience

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|--|------------------------|----------------------------|
| Fullstack Developer | Entropy Systems | May 2023 - Present |
| <ul style="list-style-type: none">Architected and delivered SaaS products for ISO standards auditing—including client-facing applications, internal dashboards, and AI-powered tooling—as part of a small engineering team (3–5 engineers).Built full-stack features in TypeScript using React, Next.js, and Vite on the frontend with Fastify and Node.js services on the backend.Integrated LLMs via AWS Bedrock and designed agent orchestration pipelines for document processing, conversational assistants, and automated business workflows.Deployed and operated distributed services across AWS (Lambda, EC2, S3, RDS, SQS/EventBridge) and GCP (Cloud Run, Cloud Functions), applying event-driven and serverless patterns.Containerized services with Docker and maintained CI/CD pipelines for reliable, repeatable deployments across cloud environments. | | |
| Frontend Engineer | Vintra, Inc. | Sep 2022 - May 2023 |
| <ul style="list-style-type: none">Led migration of a video analytics platform from legacy JavaScript to a strongly-typed TypeScript codebase, reducing runtime errors and improving developer confidence.Introduced Vite and esbuild into the build pipeline, cutting build times and accelerating development iteration speed.Partnered with product and design teams to deliver consistent UI behavior across a complex, data-rich surveillance interface. | | |
| UI Development Lead | BITA GmbH | Apr 2022 - Sep 2022 |
| <ul style="list-style-type: none">Led a front-end team (2 FTE) building index and ETF analytics tools, setting engineering standards and driving delivery quality.Implemented shift-left practices—code reviews, linting, and automated testing—to catch defects early in the development cycle.Drove front-end scalability across multiple web applications, establishing shared component patterns and consistent architecture.Contributed across the stack, delivering UI features in TypeScript and supporting backend API work in Python. | | |
| Full-Stack Engineer | BITA GmbH | Jul 2021 - Apr 2022 |
| <ul style="list-style-type: none">Built and maintained financial applications for index construction and ETF analytics across frontend, backend, and data layers.Designed asynchronous event-processing workflows using RabbitMQ to handle data ingestion reliably at scale.Developed REST APIs and backend services in Python and Node.js, backed by PostgreSQL and MongoDB. | | |
| Application Developer | Intelix Synergy | Mar 2020 - Jul 2021 |
| <ul style="list-style-type: none">Built full-stack internal systems for registration, traceability, and collaborator lifecycle management using Python and Node.js.Developed front-end interfaces and backend services to support business operations and data tracking workflows.Deployed and operated services on Linux servers with NGINX as a reverse proxy. | | |
| Software Developer (Freelance) | Freelance | May 2018 - Dec 2019 |
| <ul style="list-style-type: none">Built Python automation tools for data processing, including search and filtering workflows over large Excel datasets.Delivered custom scripts and utilities to streamline repetitive business operations for clients. | | |

Education

Skills

Programming Languages:  TypeScript,  JavaScript,  Python

Architecture & System Design: Distributed systems, Microservices, Asynchronous workflows, System design fundamentals, API design (REST)

Cloud & Infrastructure: AWS (Bedrock, Lambda, EC2, S3, RDS, SQS, EventBridge, ElastiCache, IAM, CloudWatch), GCP (Cloud Run, Cloud Functions)

Databases & Messaging: PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ

Backend: Node.js, Fastify, Express.js, FastAPI, Flask

DevOps & Tooling: Docker, CI/CD, NGINX, PM2, Git (GitHub/GitLab), Linux administration

Frontend: React, Next.js, Vite, esbuild

Collaboration: Jira, ClickUp, Trello, Slack, Discord

Certifications & Learning

- Grokking the Modern System Design Interview — Educative.io
- Grokking the Advanced System Design Interview — Educative.io
- Grokking the System Design Interview — Educative.io
- Architecting on AWS — AWS Skill Builder
- AWS Cloud Practitioner Essentials — AWS Skill Builder
- Google Cloud Fundamentals — Google Cloud Skills Boost
- Designing Distributed Systems — Google Cloud Skills Boost
- Neural Networks and Deep Learning — DeepLearning.ai
- AI For Everyone — DeepLearning.ai
- Introduction to Artificial Intelligence — Coursera
- Software Security — Coursera
- Python Data Structures — Coursera
- Backend Development with NodeJS — Platzi (School of Web Development)